24. If $p \sec 50^\circ = \tan 50^\circ$ then p is

(a) $\cos 50^\circ$ (b) $\sin 50^\circ$

(c) $\tan 50^\circ$ (d) $\sec 50^\circ$

25. $\left(\frac{\cos x}{\operatorname{cosec} x}\right) - \sqrt{1-\sin^2 x}\sqrt{1-\cos^2 x}$ is

(a) $\cos^2 x - \sin^2 x$ (b) $\sin^2 x - \cos^2 x$

(c) 1 (d) 0

Miscellaneous Problems

1. Prove that $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$

2. Prove that $\sqrt{3}\ \operatorname{cosec} 20^\circ - \sin 20^\circ = 4$

3. Prove that $\cot 4x(\sin 5x + \sin 3x) = \cot x(\sin 5x - \sin 3x)$.

4. If $\tan x = \frac{3}{4}$, $\pi < x < \frac{3\pi}{2}$ then find the value of $\sin\frac{x}{2}$ and $\cos\frac{x}{2}$

5. Prove that $2\sin^2\frac{3\pi}{4} + 2\cos^2\frac{\pi}{4} + 4\sec^2\frac{3\pi}{4} = 10$

6. Find the value of (i) $\sin 75^\circ$
(ii) $\tan 15^\circ$

7. If $\sin A = \frac{1}{3}, \sin B = \frac{1}{4}$ then find the value of $\sin(A+B)$, where A and B are acute angles.

8. Show that $\cos^{-1}\left(\frac{12}{13}\right) + \sin^{-1}\left(\frac{3}{5}\right) = \sin^{-1}\left(\frac{56}{65}\right)$

9. If $\cos(\alpha+\beta) = \frac{4}{5}$ and $\sin(\alpha-\beta) = \frac{5}{13}$ where $\alpha+\beta$ and $\alpha-\beta$ are acute, then find $\tan 2\alpha$

10. Express $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right), 0 < x < \pi$ in the simplest form.

Summary

- If in a circle of radius r, an arc of length l subtends an angle of θ radians, then $l = r\theta$
- $\sin^2 x + \cos^2 x = 1$
- $\tan^2 x + 1 = \sec^2 x$
- $\cot^2 x + 1 = \operatorname{cosec}^2 x$
- $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
- $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$
- $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$
- $\cos(\alpha-\beta) = \cos\alpha\cdot\cos\beta + \sin\alpha\sin\beta$
- $\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$
- $\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$
- $\sin 2\alpha = 2\sin\alpha\cos\alpha$
- $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 1 - 2\sin^2\alpha = 2\cos^2\alpha - 1$
- $\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$
- $\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha$
- $\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha$
- $\tan 3\alpha = \frac{3\tan\alpha - \tan^3\alpha}{1 - 3\tan^2\alpha}$
- $\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$
- $\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$
- $\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$
- $\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$
- $\sin x\cos y = \frac{1}{2}[\sin(x+y) + \sin(x-y)]$
- $\cos x\sin y = \frac{1}{2}[\sin(x+y) - \sin(x-y)]$
- $\cos x\cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)]$
- $\sin x\sin y = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$

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